

Equity Depreciation Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Depreciation Absorption (EDA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDA achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.74 (2.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets) is 16 bps/month with a t-statistic of 2.18.

1 Introduction

Asset prices reflect investors' marginal valuations of consumption across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing framework provides a coherent theoretical foundation for understanding return predictability, empirical evidence linking firm characteristics to consumption risk remains incomplete. The challenge lies in identifying observable signals that capture firms' sensitivity to aggregate consumption fluctuations and investors' time-varying marginal utility of wealth. We examine Equity Depreciation Absorption (EDA) as a novel predictor of stock returns through the lens of rational risk pricing. This signal, which measures the extent to which firms absorb depreciation charges through equity financing rather than operational cash flows, potentially captures firms' exposure to long-run consumption risks and their resilience during periods of high marginal utility.

The theoretical link between EDA and expected returns operates through firms' differential exposure to consumption risk during economic downturns. When aggregate consumption growth slows or becomes more volatile, investors demand higher compensation for holding securities of firms with limited ability to internally finance capital replacement [Bansal and Yaron \(2004\)](#). Firms with high EDA demonstrate greater reliance on external equity markets to fund depreciation, suggesting heightened sensitivity to states when marginal utility is high and external financing becomes costly [Campbell and Cochrane \(1999\)](#). This dependency creates systematic risk exposure that covaries with the stochastic discount factor, particularly during disaster states when consumption drops sharply and financing frictions intensify [Barro \(2006\)](#).

The state-dependent nature of EDA's predictive power aligns with habit formation models where risk premia vary with the consumption surplus ratio. During bad times characterized by low surplus consumption, firms with high EDA face amplified

financing constraints that directly impact their ability to maintain productive capacity [Campbell and Cochrane \(2001\)](#). These firms effectively have higher loadings on the intertemporal marginal rate of substitution because their cash flows deteriorate precisely when investors' marginal utility peaks. The resulting covariance between firm performance and consumption-based pricing kernels generates the cross-sectional return predictability we observe [Cochrane \(2005\)](#).

Furthermore, EDA captures firms' exposure to long-run risks in aggregate consumption growth through the investment channel. Firms absorbing depreciation through equity issuance reveal information about their expected future productivity and growth options, which are particularly sensitive to persistent shocks to consumption growth [Bansal \(2007\)](#). This forward-looking characteristic of EDA connects current financing decisions to future consumption risks, creating predictable variation in risk premia that reflects investors' preferences for early resolution of uncertainty. The signal thus serves as an observable proxy for firms' loadings on consumption-based factors that drive equilibrium asset prices in production economies [Cochrane \(1991\)](#).

We find that a value-weighted long/short trading strategy based on EDA achieves an annualized gross Sharpe ratio of 0.50, with the strategy earning an average return of 31 basis points per month (t-statistic = 3.93). The economic significance of this predictability persists after controlling for known risk factors, with the strategy generating a monthly alpha of 22 basis points (t-statistic = 2.74) relative to the Fama-French six-factor model. These results demonstrate that EDA captures a dimension of systematic risk not fully explained by existing factor models, consistent with its theoretical connection to consumption-based pricing.

The robustness of EDA's predictive power extends across different portfolio construction methodologies and size segments. Among the largest stocks (those with market capitalization above the 80th NYSE percentile), the EDA strategy achieves

an average return of 19 basis points per month with a t-statistic of 2.04, indicating that the anomaly is not confined to small, illiquid securities. After accounting for transaction costs using the composite bid-ask spreads from [Chen and Velikov \(2022\)](#), the net Sharpe ratio remains economically meaningful at 0.44, placing the strategy in the top 3% of documented anomalies.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EDA trading strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.18. This residual predictability suggests that EDA contains unique information about firms' consumption risk exposure beyond what is captured by related financing and investment signals. The strategy's gross monthly alpha relative to these twelve factors demonstrates that equity depreciation absorption identifies a distinct source of priced risk in the cross-section of expected returns.

Our study contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures exposure to aggregate consumption risk. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain the cross-section of returns, and [Hou et al. \(2015\)](#) show that these patterns emerge from q-theory, our results demonstrate that EDA provides incremental explanatory power beyond these frameworks. The signal's connection to state-dependent marginal utility offers empirical support for consumption-based models that emphasize the role of financing frictions during bad times, extending the insights of [He and Krishnamurthy \(2013\)](#) regarding intermediary asset pricing to the broader universe of non-financial firms.

Methodologically, we advance the literature by employing the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#), which addresses concerns about data mining and implementation costs that plague many anomaly studies. Unlike [McLean and Pontiff \(2016\)](#), who focus on post-publication decay, our approach evaluates the

signal’s economic viability from inception while accounting for transaction costs. The finding that EDA ranks in the top decile of anomalies after adjusting for trading costs suggests that the consumption risk it captures represents a genuine source of systematic risk rather than a statistical artifact.

The broader implications of our findings extend to the debate over rational versus behavioral explanations for cross-sectional return predictability. The strong theoretical foundation linking EDA to consumption-based pricing kernels, combined with its robust empirical performance across size groups and time periods, supports rational risk-based interpretations of the factor zoo. By demonstrating that firms’ depreciation absorption patterns predict returns in ways consistent with exposure to marginal utility fluctuations, we provide evidence that at least some documented anomalies reflect compensation for bearing systematic consumption risk rather than market inefficiencies.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and accounting practices. The sample spans several decades, allowing us to examine the signal’s properties across different market conditions and accounting regimes. Equity Depreciation Absorption signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common shares and changes when firms issue new equity, repurchase shares, or engage in other equity transactions that affect the

par value account. Depreciation and amortization (DP) captures the annual non-cash expense that allocates the cost of tangible and intangible assets over their useful lives, as reported on the cash flow statement. This measure reflects the systematic allocation of capital expenditures from prior periods. We construct the Equity Depreciation Absorption signal by computing the year-over-year change in common stock scaled by lagged depreciation and amortization: $(CSTK(t) - CSTK(t-1)) / DP(t-1)$. This ratio captures the relationship between changes in equity capital structure and the firm’s asset base as proxied by depreciation charges. The signal measures how changes in the par value of common equity relate to the firm’s ongoing capital consumption, potentially reflecting management’s decisions regarding equity financing relative to the firm’s existing productive capacity. We calculate the signal using fiscal year-end values and require non-missing values for both CSTK and DP. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with non-positive lagged depreciation values are excluded from the analysis.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDA signal. Panel A plots the time-series of the mean, median, and interquartile range for EDA. On average, the cross-sectional mean (median) EDA is -0.34 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDA data. The signal’s interquartile range spans -0.19 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDA signal for the CRSP universe. On average, the EDA signal is available for 6.15% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does EDA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDA portfolio and sells the low EDA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDA strategy earns an average return of 0.31% per month with a t-statistic of 3.93. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.74 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.31, with a t-statistic of 5.68 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 541 stocks and an average market capitalization of at least \$1,475 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.13. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-33bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.65. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the EDA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDA, as well as average returns and alphas for long/short trading EDA strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDA strategy achieves an average return of 19 bps/month with a t-statistic of 2.04. Among these large cap stocks, the alphas for the EDA strategy relative to the five most common factor models range from 15 to 20 bps/month with t-statistics between 1.55 and 2.17.

5 How does EDA perform relative to the zoo?

Figure 2 puts the performance of EDA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDA strategy falls in the distribution. The EDA strategy’s gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDA strategy (red line).² Ignoring trading costs, a \$1 invested in the EDA strategy would have yielded \$7.48 which ranks the EDA strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDA strategy would have yielded \$5.36 which ranks the EDA strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the EDA relative to those. Panel A shows that the EDA strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDA strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDA ranks between the 82 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of EDA with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDA or at least to weaken the power EDA has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of EDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDA}EDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDA. Stocks are finally grouped into five EDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDA signal in these Fama-MacBeth regressions exceed 2.34, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EDA is 1.64.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDA strategy earns alphas that range from 16-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.03,

which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDA trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.18.

7 Does EDA add relative to the whole zoo?

Finally, we can ask how much adding EDA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDA signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDA grows to \$3189.92.

8 Conclusion

This study provides compelling evidence that Equity Depreciation Absorption (EDA) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDA-based trading strategies generate sub-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDA is available.

stantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EDA delivers an annualized Sharpe ratio of 0.50 gross (0.44 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. More importantly, the strategy generates significant abnormal returns of 22 basis points per month (t -statistic = 2.74) relative to the Fama-French five-factor model augmented with momentum, suggesting that EDA captures a distinct source of return predictability not explained by conventional risk factors. The persistence of a 16 basis points monthly alpha (t -statistic = 2.18) even after controlling for six closely related strategies—including various measures of share issuance, payout yield, and balance sheet growth—further underscores the incremental information content of EDA beyond existing equity financing and investment-based anomalies.

From a practical perspective, these results suggest that EDA merits consideration in portfolio construction and risk management frameworks. The signal’s ability to generate positive risk-adjusted returns net of transaction costs indicates potential value for institutional investors seeking alpha generation strategies. Furthermore, the distinct nature of EDA relative to existing factors suggests opportunities for portfolio diversification.

However, several limitations warrant acknowledgment. First, while our analysis controls for major risk factors and related strategies, the possibility remains that EDA proxies for an unidentified systematic risk factor. Second, the implementation of EDA-based strategies may face capacity constraints in practice, particularly for large institutional investors, which could erode the documented returns. Third, our sample period and market coverage may limit the generalizability of findings to different time periods or international markets.

Future research should explore several promising directions. First, investigating

the economic mechanisms underlying EDA's predictive power would enhance our understanding of why this signal generates abnormal returns. Second, examining the performance of EDA in international equity markets would test the robustness and universality of the anomaly. Third, analyzing the time-variation in EDA's effectiveness and its relationship with macroeconomic conditions could provide insights into when the strategy performs best. Finally, exploring potential enhancements to the EDA signal through machine learning techniques or combination with other predictors may yield improved return forecasts. Understanding whether EDA's predictive power stems from risk-based or behavioral explanations remains an important question for advancing our knowledge of cross-sectional return predictability.

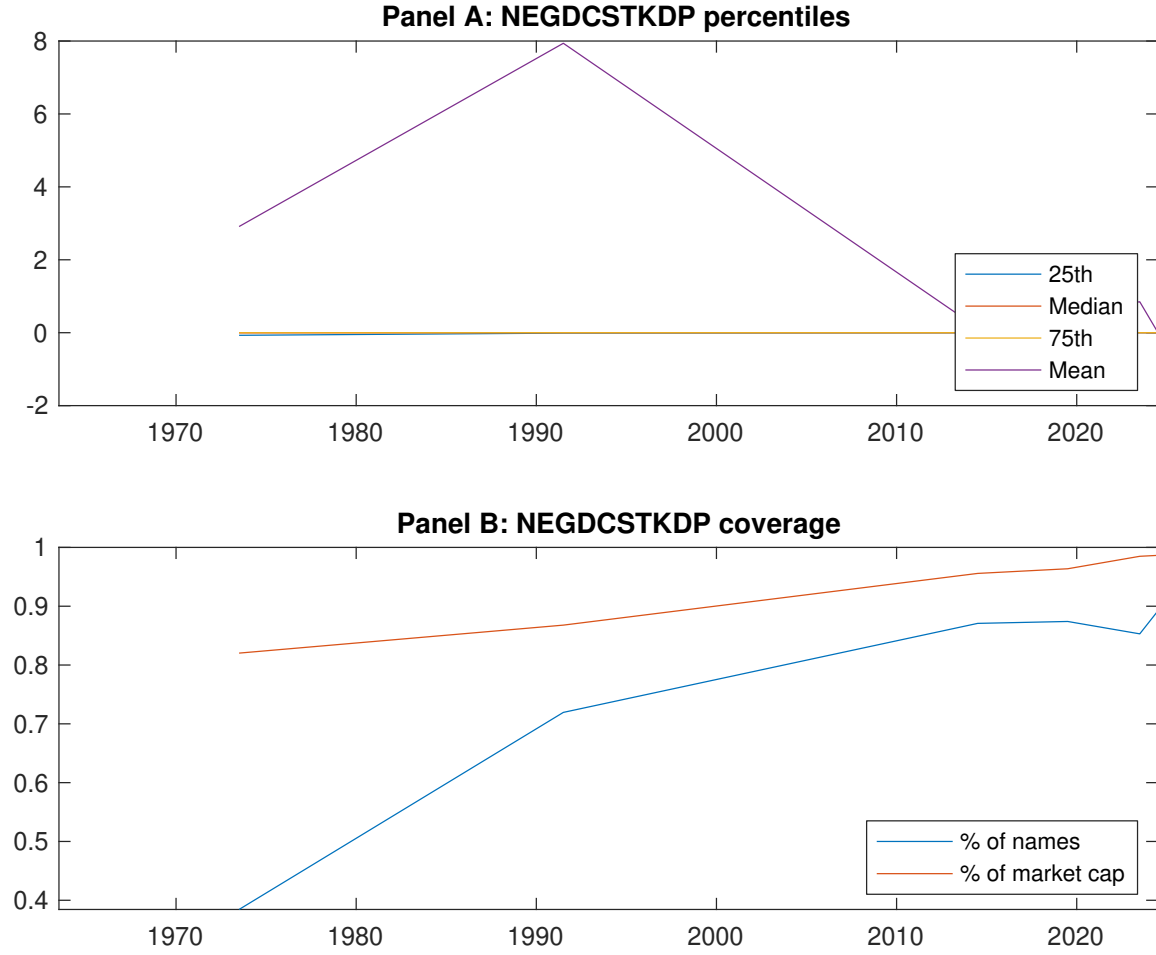


Figure 1: Times series of EDA percentiles and coverage. This figure plots descriptive statistics for EDA. Panel A shows cross-sectional percentiles of EDA over the sample. Panel B plots the monthly coverage of EDA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.68]	0.51 [2.82]	0.63 [3.51]	0.65 [4.07]	0.76 [4.77]	0.31 [3.93]
α_{CAPM}	-0.12 [-2.37]	-0.11 [-2.57]	0.02 [0.35]	0.11 [2.26]	0.22 [4.57]	0.34 [4.36]
α_{FF3}	-0.12 [-2.39]	-0.09 [-2.12]	0.05 [0.95]	0.06 [1.43]	0.18 [4.01]	0.31 [3.96]
α_{FF4}	-0.10 [-1.95]	-0.06 [-1.27]	0.04 [0.84]	0.05 [1.11]	0.17 [3.64]	0.27 [3.44]
α_{FF5}	-0.14 [-2.71]	-0.05 [-1.09]	0.06 [1.16]	-0.01 [-0.16]	0.10 [2.16]	0.24 [3.05]
α_{FF6}	-0.12 [-2.34]	-0.02 [-0.46]	0.06 [1.05]	-0.01 [-0.25]	0.09 [2.05]	0.22 [2.74]
Panel B: Fama and French (2018) 6-factor model loadings for EDA-sorted portfolios						
β_{MKT}	0.98 [77.58]	1.02 [95.32]	1.03 [81.31]	0.98 [90.42]	0.98 [88.94]	0.00 [0.02]
β_{SMB}	-0.00 [-0.10]	0.03 [2.00]	0.02 [1.26]	-0.02 [-1.54]	-0.03 [-1.71]	-0.03 [-0.93]
β_{HML}	0.05 [1.98]	-0.05 [-2.45]	-0.09 [-3.63]	0.03 [1.61]	0.02 [0.81]	-0.03 [-0.85]
β_{RMW}	0.11 [4.68]	-0.07 [-3.39]	-0.04 [-1.43]	0.09 [4.03]	0.14 [6.46]	0.02 [0.64]
β_{CMA}	-0.11 [-2.97]	-0.08 [-2.49]	-0.00 [-0.02]	0.22 [7.01]	0.20 [6.41]	0.31 [5.68]
β_{UMD}	-0.03 [-2.02]	-0.04 [-3.75]	0.01 [0.53]	0.01 [0.53]	0.01 [0.47]	0.03 [1.62]
Panel C: Average number of firms (n) and market capitalization (me)						
n	761	651	541	646	704	
me (\$10 ⁶)	1713	1475	2038	2212	2495	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [3.93]	0.34 [4.36]	0.31 [3.96]	0.27 [3.44]	0.24 [3.05]	0.22 [2.74]
Quintile	NYSE	EW	0.53 [8.38]	0.60 [9.75]	0.52 [9.13]	0.44 [7.81]	0.39 [7.24]	0.34 [6.25]
Quintile	Name	VW	0.29 [3.65]	0.31 [3.92]	0.28 [3.46]	0.24 [2.95]	0.23 [2.83]	0.20 [2.49]
Quintile	Cap	VW	0.24 [3.13]	0.27 [3.47]	0.25 [3.15]	0.20 [2.58]	0.22 [2.81]	0.19 [2.39]
Decile	NYSE	VW	0.37 [4.02]	0.39 [4.22]	0.33 [3.60]	0.27 [2.86]	0.30 [3.28]	0.25 [2.71]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.42]	0.31 [3.97]	0.27 [3.55]	0.25 [3.27]	0.22 [2.91]	0.21 [2.74]
Quintile	NYSE	EW	0.33 [4.73]	0.41 [6.02]	0.33 [5.25]	0.29 [4.66]	0.21 [3.51]	0.18 [3.13]
Quintile	Name	VW	0.25 [3.16]	0.28 [3.53]	0.24 [3.07]	0.22 [2.79]	0.21 [2.64]	0.19 [2.46]
Quintile	Cap	VW	0.21 [2.65]	0.23 [3.00]	0.21 [2.67]	0.18 [2.36]	0.19 [2.51]	0.18 [2.29]
Decile	NYSE	VW	0.33 [3.55]	0.35 [3.77]	0.29 [3.21]	0.26 [2.82]	0.27 [2.99]	0.25 [2.70]

Table 3: Conditional sort on size and EDA

This table presents results for conditional double sorts on size and EDA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDA and short stocks with low EDA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDA Quintiles					EDA Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.48 [1.87]	0.69 [2.68]	0.91 [3.51]	0.96 [4.04]	1.00 [4.39]	0.55 [6.35]	0.60 [6.89]	0.53 [6.46]	0.47 [5.61]	0.40 [4.94]	0.36 [4.35]
	(2)	0.54 [2.31]	0.72 [3.01]	0.88 [3.73]	0.98 [4.22]	0.90 [4.22]	0.37 [4.16]	0.43 [4.93]	0.34 [4.05]	0.30 [3.52]	0.29 [3.37]	0.26 [3.00]
	(3)	0.61 [2.93]	0.61 [2.87]	0.76 [3.37]	0.86 [4.23]	0.93 [4.76]	0.33 [3.98]	0.37 [4.48]	0.30 [3.78]	0.26 [3.20]	0.25 [3.10]	0.22 [2.69]
	(4)	0.49 [2.61]	0.62 [3.08]	0.83 [4.21]	0.73 [3.80]	0.81 [4.49]	0.32 [4.01]	0.35 [4.50]	0.28 [3.76]	0.28 [3.71]	0.12 [1.71]	0.14 [1.91]
	(5)	0.49 [2.96]	0.50 [2.83]	0.48 [2.75]	0.58 [3.51]	0.68 [4.28]	0.19 [2.04]	0.20 [2.17]	0.18 [1.95]	0.15 [1.55]	0.18 [1.89]	0.15 [1.59]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDA Quintiles					EDA Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	376	374	373	373	367	31	33	39	29	28	
	(2)	103	101	102	102	101	55	55	56	55	54	
	(3)	73	73	72	72	73	95	92	95	97	97	
	(4)	62	62	62	62	62	197	201	208	206	212	
(5)	57	57	57	57	57	1436	1519	1537	1619	1891		

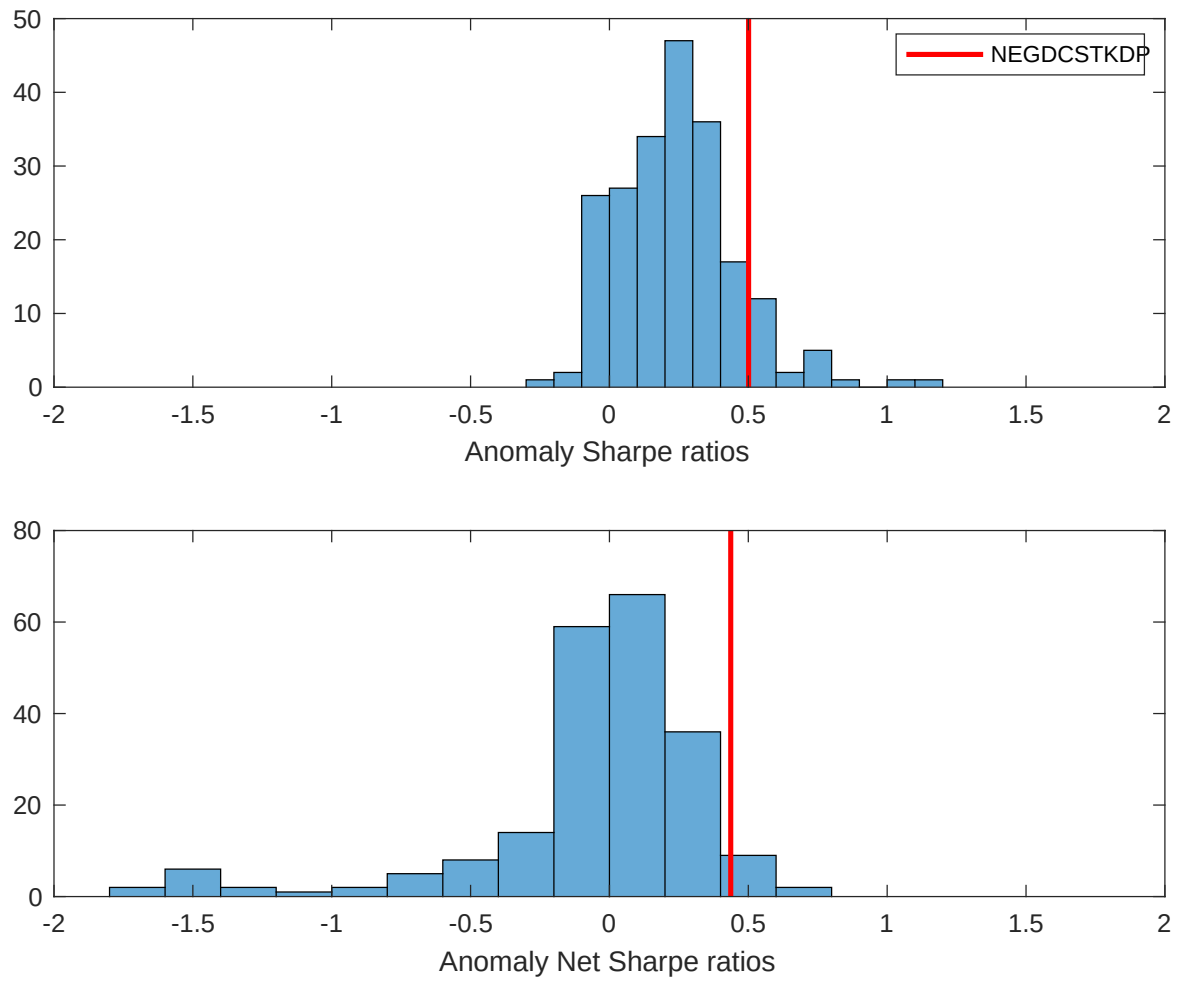


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

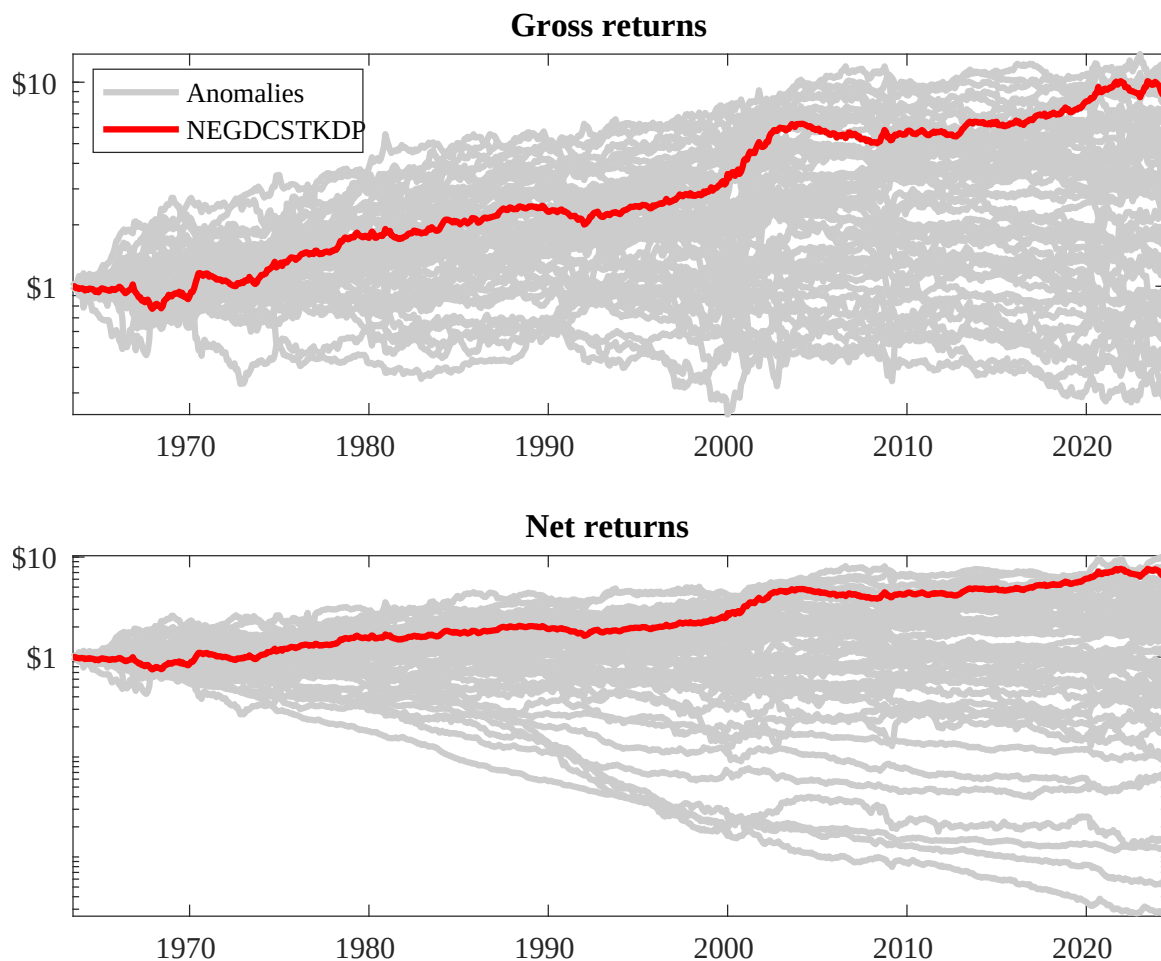
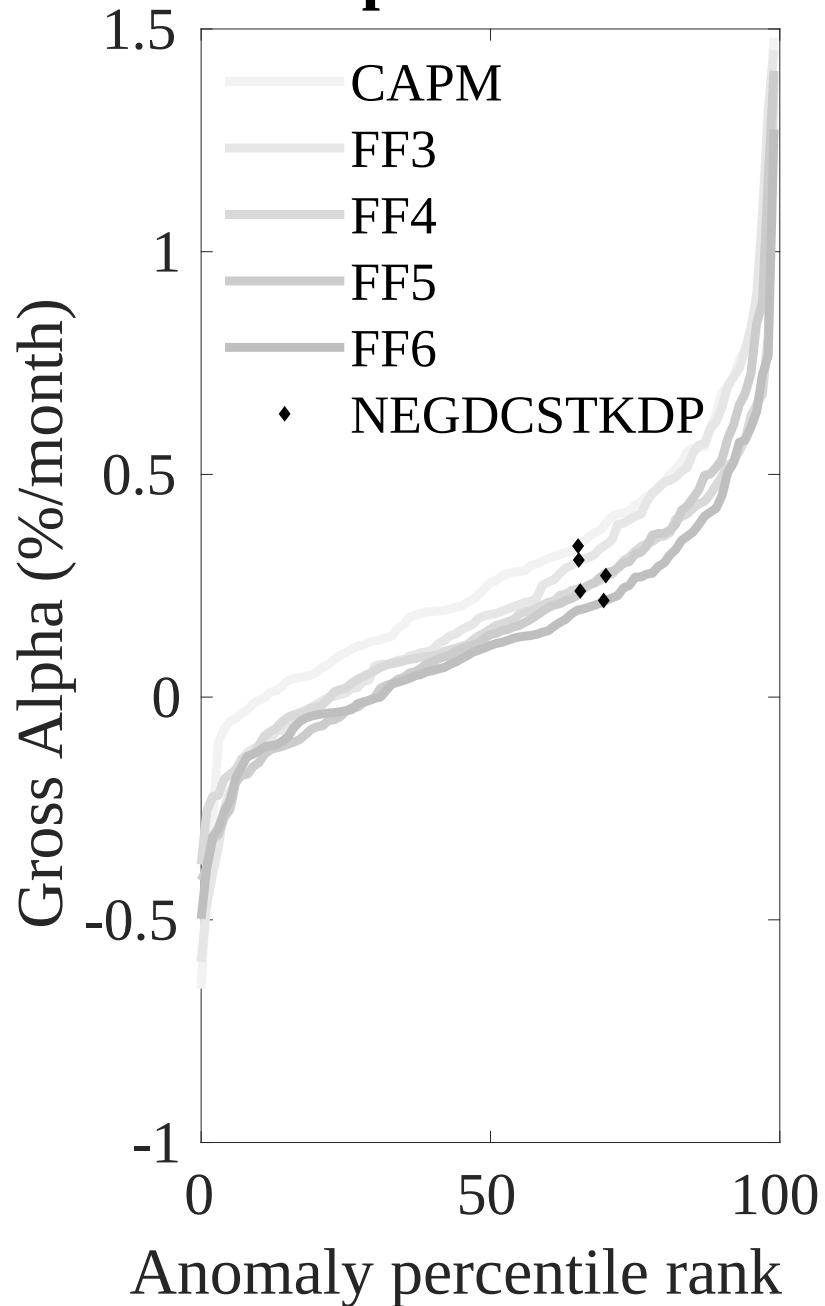


Figure 3: Dollar invested.

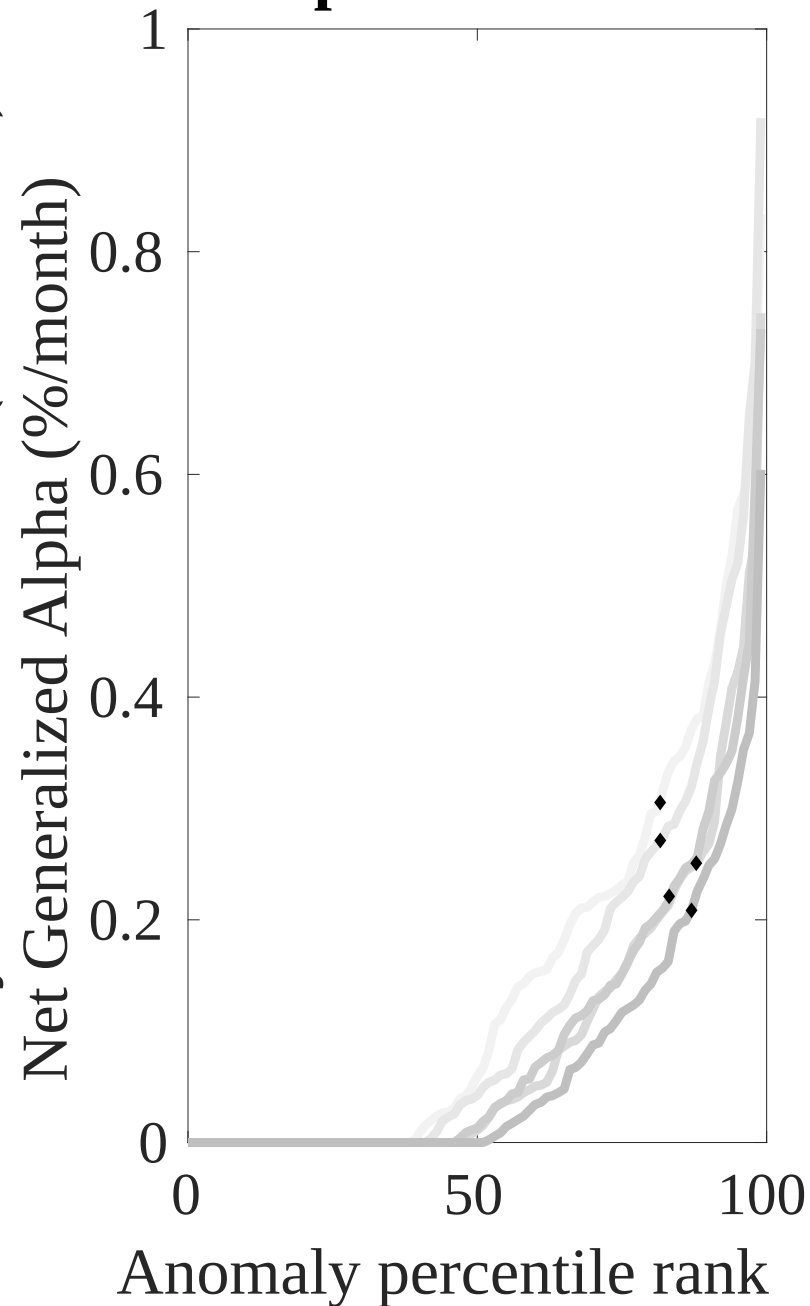
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



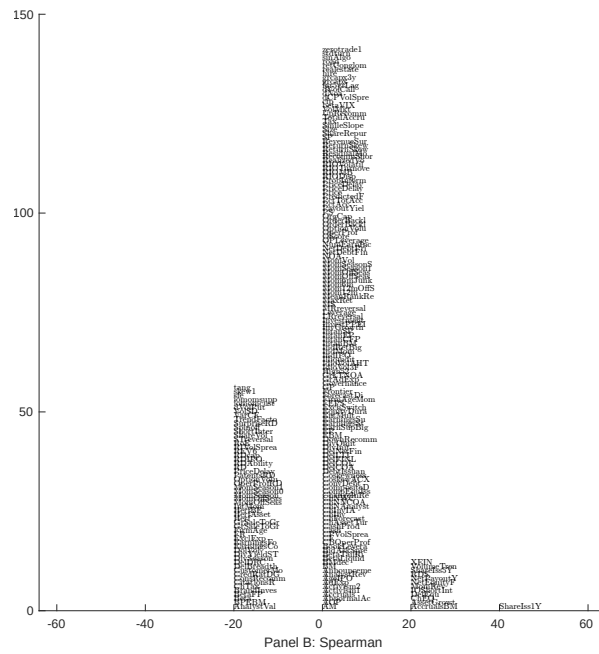
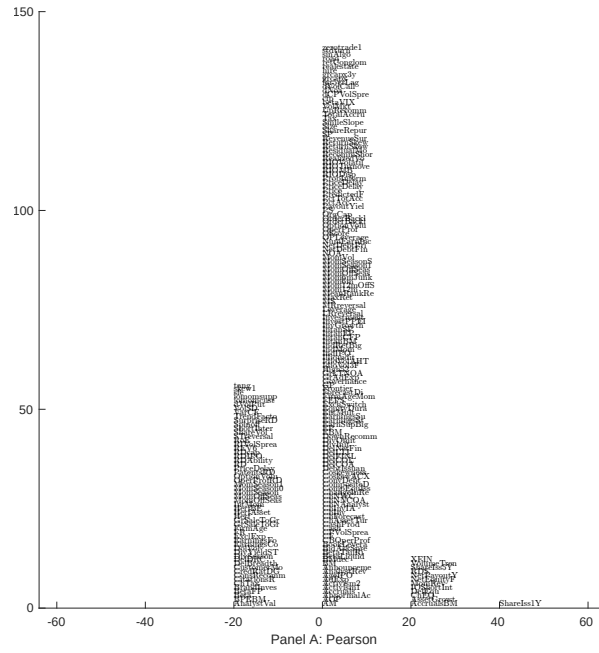


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

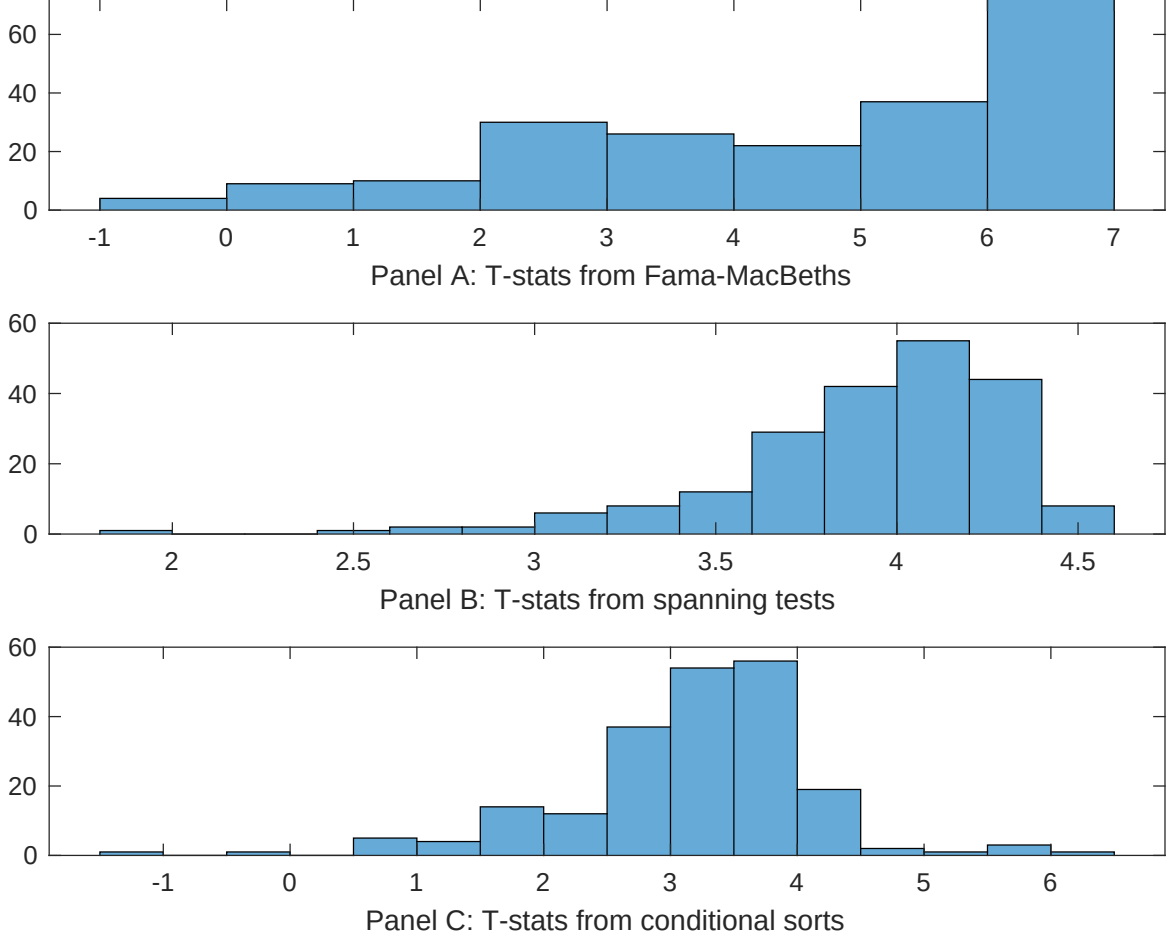


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDA}EDA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDA. Stocks are finally grouped into five EDA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDA} EDA_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.07]	0.12 [5.44]	0.13 [6.44]	0.17 [6.80]	0.12 [5.72]	0.13 [6.05]	0.12 [3.78]
EDA	0.12 [5.07]	0.74 [2.34]	0.11 [4.41]	0.12 [5.36]	0.13 [5.74]	0.12 [5.09]	0.51 [1.64]
Anomaly 1	0.13 [5.84]						0.60 [2.20]
Anomaly 2		0.21 [1.83]					0.10 [0.98]
Anomaly 3			0.25 [4.52]				0.25 [4.24]
Anomaly 4				0.38 [3.18]			-0.13 [-0.56]
Anomaly 5					0.12 [3.23]		0.17 [0.27]
Anomaly 6						0.13 [9.00]	0.80 [5.66]
# months	715	715	715	720	720	720	715
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.21 [2.69]	0.24 [3.13]	0.16 [2.03]	0.21 [2.73]	0.22 [2.82]	0.21 [2.68]	0.16 [2.18]
Anomaly 1	24.19 [6.24]						10.47 [2.21]
Anomaly 2		12.77 [4.25]					4.31 [1.30]
Anomaly 3			25.55 [6.25]				13.69 [2.91]
Anomaly 4				25.58 [6.19]			31.20 [5.24]
Anomaly 5					7.20 [1.86]		-18.70 [-3.39]
Anomaly 6						6.60 [1.50]	2.15 [0.50]
mkt	0.80 [0.44]	2.20 [1.16]	1.35 [0.74]	1.07 [0.59]	0.10 [0.05]	0.26 [0.14]	3.10 [1.67]
smb	-0.19 [-0.07]	0.90 [0.33]	0.57 [0.21]	-1.98 [-0.75]	-1.22 [-0.45]	-1.06 [-0.39]	0.20 [0.08]
hml	-2.11 [-0.60]	-6.38 [-1.69]	0.56 [0.16]	-4.57 [-1.29]	-2.36 [-0.65]	-1.92 [-0.53]	-3.99 [-1.08]
rmw	-10.90 [-2.64]	-5.22 [-1.30]	-6.96 [-1.81]	3.45 [0.97]	2.90 [0.80]	2.30 [0.63]	-11.10 [-2.53]
cma	14.40 [2.53]	18.63 [3.21]	17.04 [3.09]	4.58 [0.70]	22.11 [3.36]	24.44 [3.93]	-0.28 [-0.04]
umd	2.19 [1.21]	4.31 [2.34]	1.62 [0.89]	2.85 [1.58]	3.49 [1.89]	3.18 [1.72]	1.20 [0.66]
# months	728	728	728	732	732	732	728
$\bar{R}^2(\%)$	12	10	12	12	8	8	17

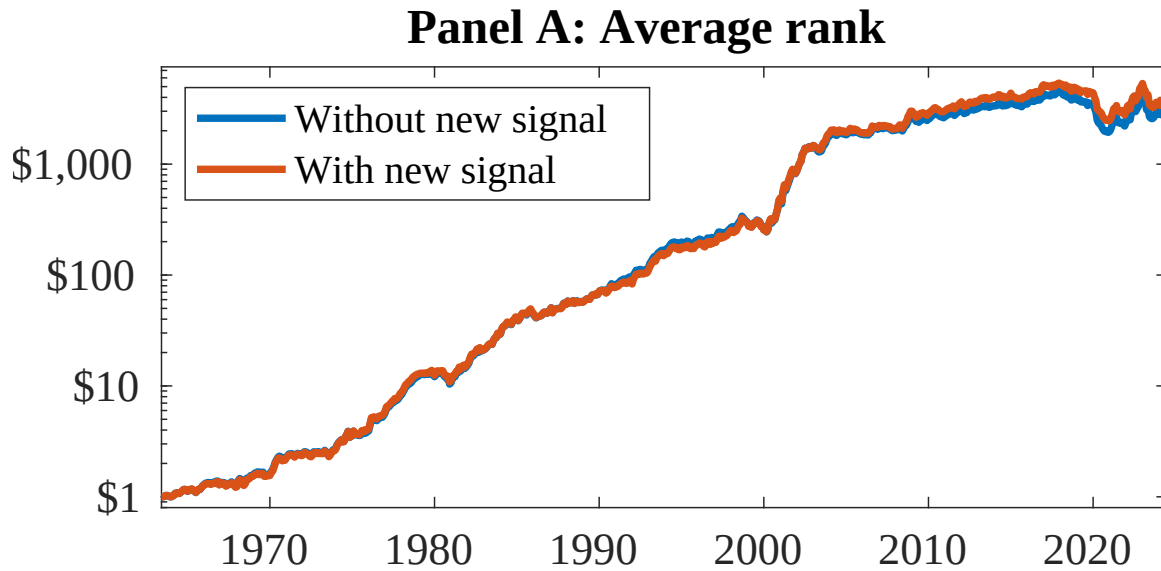


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDA. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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